

JUYEON MA

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Languages: English (IH), Korean (Fluent)

SUMMARY

Software Developer with hands-on experience in C/C++-based industrial systems, real-time data acquisition, and communication protocols including TCP/IP and serial communication. Experienced in EV battery testing environments and embedded system integration. Skilled in debugging, protocol-level issue resolution, and ensuring system reliability.

TECHNICAL SKILLS

Languages: Python, C/C++

Tools & Frameworks: Visual Studio, Django, SQLite, MFC, VSCode

Core Skills: Socket Programming, Embedded Systems, Real-Time Data Processing, Web Development, Database Integration, AI Model Integration

WORK EXPERIENCE

Software Developer — APRO

Anyang, South Korea | Jul 2024 – Sep 2024

- Developed EV battery testing software in C/C++ supporting real-time acquisition and analysis of voltage, current, and temperature data.
- Contributed to communication architecture using serial and TCP socket protocols; implemented packet structures and error-handling logic.
- Performed on-site testing with EV batteries, including charge/discharge and protection circuit validation.
- Identified and resolved issues such as communication errors, missing data frames, and UI delays, improving system responsiveness.
- Supported partial automation of repetitive test procedures to reduce test time and improve consistency.

PROJECT EXPERIENCE

1. AI-Based Music Generation Platform — KT AIVLE School

Frontend & Backend Contributor | Dec 2023 – Jan 2024

- Developed main page, user registration page, and bulletin board UI using Django, HTML/CSS, and JavaScript.
- Integrated frontend pages with Django backend and connected SQLite database to store user posts and comments.
- Collaborated with backend and AI teams to support post/comment features and integrate MusicGEN model outputs into the web interface.
- Contributed to delivering a functional AI-based music generation service recognized with the Practical Award.

2. Helmet Detection System for Electric Scooters — Capstone Project

Developer (Socket Communication) | Mar 2021 – Jun 2021

- Implemented bidirectional TCP socket communication between helmet (server) and scooter (client) using Python and Raspberry Pi.
- Integrated ultrasonic, pressure, and PIR sensors to detect rider presence and verify helmet usage.
- Improved real-time response and safety by enabling scooter operation only when helmet detection was successful.

3. COVID-19 Integrated Management System — Database Project

Database Developer | Nov 2021 – Dec 2021

- Designed pharmaceutical company database and implemented SQL-based data retrieval and modification features.
- Built vaccine inventory and order management system using MySQL.

Education

B.S. Information & Communication Engineering (Mar 2020 - Feb 2023)

Yeungnam University, South Korea

GPA: 3.56/4.5

A.A.S. Electronic Information & Communication

Yeungjin College, South Korea

GPA: 3.58/4.5

CERTIFICATION

Computer Utilization Level 2 (2023)

AWARDS

AIVLE Big Project Practical Award (2024)